

AMANI MUTHYALA

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Career Objective

Third-year Electronics and Communication Engineering student with a strong interest in Embedded Systems and IoT. Actively seeking an internship opportunity to gain hands-on industry experience and enhance technical skills in a professional environment.

Education

Rajiv Gandhi University of Knowledge and Technologies, IIIT RK Valley	<i>2023 – Present</i>
Bachelor of Technology (B.Tech) in Electronics and Communication Engineering	CGPA: 7.52
Rajiv Gandhi University of Knowledge and Technologies, IIIT RK Valley	<i>2021 – 2023</i>
Pre-University Course (PUC)	CGPA: 9.30
Sri Krishna Devaraya EM High School	<i>2020 – 2021</i>
Secondary School Certificate (SSC)	CGPA: 10.0

Technical Strengths

Programming Languages	C, Embedded C, HTML, Python (Basics)
Core Domains	Analog & Digital Electronics, IoT, Microcontrollers, Sensors, Protocols
Tools & Platforms	VS Code, Arduino IDE, Tinkercad, Keil v4, Proteus
Soft Skills	Teamwork, Communication Skills, Time Management

Projects

Predictive Drain Monitoring for Resilient Cities

- Developed an embedded IoT-based predictive drain monitoring system using ESP32 and Arduino.
- Detected water level, blockage, and overflow conditions in drainage systems.
- Processed real-time sensor data using Embedded C.
- Transmitted alerts through cloud connectivity to help prevent urban flooding.

Safe Evac – Community Flood Monitoring and Early Evacuation System (Mini Project)

- Designed an embedded IoT flood monitoring system for low-lying flood-prone areas.
- Used ultrasonic and rain sensors integrated with ESP32.
- Enabled real-time cloud data transmission.
- Provided early evacuation alerts using buzzers and LEDs.

IoT-Based Home Automation System

- Built an ESP32-based IoT home automation system for wireless LED control.
- Implemented battery-powered portable operation.
- Improved user convenience and energy efficiency.

Academic Achievements

- Project shortlisted among **Top 50 projects out of 500** at the college level for Smart India Hackathon (SIH) for developing an Embedded IoT-based Predictive Drain Monitoring system.
- Completed **edX Certification in Internet of Things (IoT)** covering IoT concepts, protocols, and embedded applications.

Extra-Curricular Activities

- Volunteer at **National Service Scheme (NSS)**, actively participating in social and community activities.
- Attended **AStronics Robotics Workshop** conducted at college.
- Participated in the **Hackathon** conducted by Everest Engineering with the Predictive Drain Monitoring Embedded **IoT project for real-time urban flood and drainage monitoring**.
- Participated in **Instinct4.0 Hackathon** with the **Smart Asset Tracking Tag project**, using **smart meters for real-time asset monitoring** and management